Project Plan

PSCP Workforce Locator

Preparedness, Safety & Continuity Portal: Workforce Locator

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Course: Information Technology

Date: May 2026

1. Introduction

1.1 Project Title

Preparedness, Safety & Continuity Portal (PSCP): Workforce Locator

1.2 Project Description

The PSCP Workforce Locator is a web-based application that tracks the real-time locations of employees and displays them on an interactive map. It also shows live disaster data (like earthquakes and natural events) so administrators can quickly see if any personnel are near a hazard. The goal is to replace manual check-in methods with a simple, map-based monitoring system.

1.3 Purpose of This Document

This project plan explains what the system does, how it was built, what tools and technologies were used, and what is planned for future development. It serves as a guide to the overall project from start to finish.

1.4 Objectives

- Allow employees to report their GPS location through a web browser.
- Show all employee locations on an interactive admin map.
- Display live earthquake and natural disaster data alongside employee positions.
- Keep a complete history log of all location check-ins for auditing.
- Secure all data with login authentication and role-based access.

2. Project Scope

2.1 What the System Does

Feature	Description

User Login & Roles	Employees and Admins have separate access levels.
GPS Location Tracking	Employees submit their location via the browser's Geolocation API.
Admin Command Center	An interactive map showing all employees and live disaster markers.
Disaster Data	Live earthquake data from USGS and natural events from NASA.
Map Layers	Toggleable layers for active faults, volcanoes, and KMZ files.
Workforce Geography	A page to compare Home vs. Office employee distribution.
Audit Logs	Full history of every location submission per employee.

2.2 What is NOT Included (For Now)

- Native mobile app (iOS/Android) — planned for future.
- Automatic hazard alerts and predictions — planned for future.
- Geofencing (auto check-in when entering an area) — planned for future.

3. Tools and Technologies Used

Category	Tool / Technology	What It Does
Backend Framework	Laravel 12 (PHP 8.2+)	Handles the server-side logic, routing, and database.
Frontend Styling	Tailwind CSS v4	Makes the pages look good and responsive.

Template Engine	Blade	Laravel's built-in way to render HTML pages.
Database	MySQL	Stores user accounts and location records.
Map Library	Leaflet.js	Renders the interactive maps with markers and layers.
Charts	Chart.js	Creates the graphs on the admin dashboard.
Build Tool	Vite 7	Bundles and serves the CSS/JS files during development.
Deployment	Docker / Nixpacks	Packages the app for easy deployment.

4. System Modules

The system is divided into the following main modules:

Module 1: Authentication & User Management

- Login and registration system.
- Two roles: Employee and Admin.
- Profile management (name, office, mobile number, employee type).

Module 2: Location Tracking

- Uses the browser's Geolocation API to get GPS coordinates.
- Automatically converts coordinates to a readable address (reverse-geocoding).
- Employees choose if they're reporting from "Home" or "Office".

- API endpoint is rate-limited to 30 requests per minute for security.

Module 3: Admin Command Center

- Interactive map that plots all employee locations.
- Live earthquake and NASA event markers displayed on the same map.
- Toggleable layers: active faults, volcanoes, KMZ data.
- Dashboard stats: total users online, recent activity, distribution charts.

Module 4: Disaster Tracking

- Fetches earthquake data from the USGS API.
- Fetches natural events (wildfires, storms, etc.) from NASA EONET API.
- Renders environmental data from KMZ/KML files.

Module 5: Workforce Geography

- Separate page to analyze Home vs. Office employee distribution.
- Clicking an employee in the list highlights them on the map.
- Solo marker mode for focusing on one person.

Module 6: Audit & History

- Complete log of every location submission.
- Admins can search and filter logs by date, department, or employee.
- Data is stored permanently and cannot be edited after submission.

5. Development Timeline

The project was developed in phases:

Phase 1: Core Features (Done)

Task	Status
GPS location capture from the browser	Done

Admin dashboard with activity feed	Done
Mobile-friendly employee interface	Done
Docker setup for easy deployment	Done
Login system with admin/employee roles	Done
Reverse-geocoding (coordinates to address)	Done

Phase 2: Disaster Data Integration (Done)

Task	Status
USGS earthquake data on the map	Done
NASA natural events on the map	Done
Active fault line layer	Done
Volcano locations layer	Done
KMZ/KML file rendering	Done

Phase 3: Performance Improvements (Done)

Task	Status
Database indexing for faster queries	Done
Skeleton loading screens	Done
Optimized page transitions	Done
Merged Command Center into dashboard	Done
Synchronized map-list interactions	Done

Phase 4: Future Plans

Task	Status
Predictive alerts (warn if disaster is near employees)	Planned
Geofencing (auto check-in by area)	Planned
Native mobile app with background tracking	Planned
Push notifications for nearby hazards	Planned

6. Security Measures

The system includes the following security features:

Security Feature	How It Works
Role-Based Access	Only Admins can see the command center and employee logs. Regular employees can only submit their own location.
Rate Limiting	The location API only allows 30 requests per minute to prevent spam or abuse.
CSRF Protection	Laravel automatically protects all forms from cross-site request forgery attacks.
Password Hashing	All passwords are hashed using bcrypt before storing in the database.
Input Validation	GPS coordinates are strictly validated before saving.

7. Testing Summary

The following tests were done to make sure the system works correctly:

What Was Tested	Result
SQL Injection (trying to hack the database through input)	Safe — Laravel uses parameterized queries.
Unauthorized Access (employee trying to access admin pages)	Safe — AdminMiddleware blocks it.
API Spam (sending too many location requests)	Safe — Rate limiting is active.
Cross-Site Scripting (XSS)	Safe — Laravel escapes output by default.
Page Load Speed	Fast — Skeleton loaders and indexed queries help.

8. Problems Encountered and Solutions

Problem	Solution
GPS is inaccurate when the user is indoors.	Added guidance for users to report location outdoors or near a window.
USGS/NASA APIs sometimes time out.	Added error handling so the map still loads even if disaster data fails.
Map gets slow with many layers active.	Optimized layer rendering and added toggle controls.
Users deny browser location permission.	Added clear instructions and a fallback message explaining how to enable it.
Large KMZ files take time to load.	Used pre-processed local data files instead of parsing on every load.

9. Conclusion and Recommendations

9.1 Conclusion

The PSCP Workforce Locator successfully provides a working system for tracking employee locations in real time and displaying them alongside live disaster data. All core features (GPS tracking, admin dashboard, disaster integration, audit logs) have been completed and tested. The system is deployed and functional.

9.2 Recommendations for Future Work

- Add predictive alerts that warn admins when disasters are heading toward employees.
- Build a mobile app for easier background location tracking.
- Add geofencing so employees are automatically checked in when they arrive at the office.
- Set up automated daily database backups for disaster recovery.
- Add push notifications for real-time hazard proximity warnings.

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